Uneven Illumination Correction

Homomorphic filtering: complete explanation with examples

SIMATS Engineering | Computer Vision (ITA05) | Assessment Tool 2: Scenario-Based

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1. The scenario and the questions

> **Scenario:** A photo taken under non-uniform light has bright patches and dark patches.

- **(a)** What is the effect of uneven light on the image?
- **(b)** What causes the brightness variation?
- **(c)** Which method can fix it?
- **(d)** Why did you choose this method?
- **(e)** Write your answer step by step.

How to use this document: Part 2 explains the basic ideas first. Parts 3 to 7 answer (a) to (e) in detail with worked numbers. Part 8 is a short exam-ready answer, and Part 9 has extra viva questions.

2. Basic ideas you must know first

2.1 What is a digital image?

A digital image is a grid of tiny squares called **pixels**. In a gray image every pixel stores one number from **0 to 255**. The number 0 means black, 255 means white, and the numbers in between are shades of gray. A camera measures how much light comes back from each tiny spot of the scene and saves it as that number. So the pixel value depends on **how much light fell on the spot** and **how much of it the surface reflects**.

Gray levels from bright to dark



Figure 1: A smooth change from bright (255) to dark (about 40). Uneven light makes one photo contain a whole range like this.

2.2 Illumination and reflectance

Every pixel value is made of two parts multiplied together:

Light L(x,y) how much light falls on the spot	X	**Surface R(x,y)** how much the object reflects (the real picture)	=	**Photo I(x,y)** the pixel value we see

Figure 2: The illumination-reflectance model, I = L x R.

Reflectance R is what we want: the paper is bright (for example 0.9) and the ink is dark (for example 0.2). It changes quickly, at the edges of letters and objects. **Illumination L** is what we do not want: it changes **slowly and smoothly** across the image, from a bright area to a dark area.

2.3 What is uneven (non-uniform) illumination?

The light is not the same at every place in the scene, so L(x,y) is not constant. Some parts of the photo get a lot of light (bright patches) and some get little (dark patches).

Part of the photo	Light L	What you see
Bright patch	High (near 1.0)	Pixels near 255, details washed out
Dark patch	Low (near 0.2)	Pixels near 0 to 50, details hidden in shadow

Key property: the light changes **slowly** (low frequency) but the real picture changes **fast** (high frequency). This one fact decides which method we should use.

2.4 What is happening in this scenario, and what do we need?

- The photo has bright and dark patches. The same paper or object looks bright in one place and dark in another.
- We need to remove the slow light variation without damaging the real details (edges, text, texture).
- So we need a method that separates L from R and keeps only R. That method is **homomorphic filtering**.

3. Answer to (a): What is the effect of uneven light?

Answer: Dark areas lose detail, bright areas get washed out, the same object looks different in different places, and later steps such as thresholding, edge detection and OCR fail.

Effect	What happens	Result in the image
Lost detail in dark patches	Pixel values are squeezed near 0, so small differences are invisible.	Shadows hide text and objects.
Washed-out bright patches	Pixel values reach 255 and cannot go higher (saturation).	Details burn out to plain white.
Same surface, different gray	Paper that is really one shade shows many gray levels.	Image looks patchy and unnatural.
Wrong results in later steps	One global threshold cannot suit both bright and dark areas.	Segmentation, edge detection, OCR and matching fail.

Why it matters: in the example of Part 7, a dark letter on the bright left side has value 41, while blank paper on the dark right side has value 46. The ink on one side is as bright as the paper on the other side, so the computer cannot tell text from paper.

4. Answer to (b): What causes the brightness variation?

Answer: The light falling on the scene is not uniform. Causes are a single or one-sided light source, shadows, lens vignetting, a tilted or curved surface, and limited camera exposure range.

Cause	What happens	Result in the image
One-sided or close light source	Light falls off with distance and angle, so the side near the lamp gets more.	Smooth bright-to-dark gradient.
Shadows	A hand, phone, book or object blocks part of the light.	Dark patch with soft edges.

Cause	What happens	Result in the image
Mixed light sources	A window and a ceiling lamp light different areas differently.	Bright and dark patches in different places.
Lens vignetting	The lens passes less light at the corners than at the centre.	Dark corners.
Tilted or curved surface	The surface faces the light at different angles (curved page, folded paper).	Brightness changes along the curve.
Limited exposure range	The camera cannot record very bright and very dark areas together.	Bright areas saturate, dark areas clip.

Common point: in every case the change is in the **light L**, not in the object. The object itself (the surface R) has the same colour everywhere, and the change in L is slow and smooth.

5. Answer to (c): Which method can fix it?

Answer: Use **homomorphic filtering** (log, low-pass estimate of the light, subtract, exp).

5.1 Key words

- **Filter:** a rule that recalculates each pixel using the pixels around it, or a rule that keeps some frequencies and removes others.
- **Low frequency:** slow changes, such as smooth light. **High frequency:** fast changes, such as edges and text.
- **Logarithm (ln):** turns multiplication into addition: ln(L x R) = ln L + ln R.
- **Low-pass (Gaussian) filter:** a large blur. It keeps only slow changes, so it finds the light L.

5.2 Why does homomorphic filtering remove uneven light?

A photo is **I = L x R**. The two parts are multiplied, so an ordinary filter cannot separate them. After taking the logarithm, they are **added**: ln I = ln L + ln R. Now L is the slow part and R is the fast part, so a filter can split them. A large Gaussian blur of ln I gives ln L (the light). Subtracting it leaves ln R (the surface). Taking exp gives back R. The result looks as if the scene had been lit evenly.

5.3 Understanding with the example scan line

Figure 3 shows one scan line of 8 pixels (the full numbers are in Part 7). The red line is the photo: it falls from 230 to 46 because the light fades. The text (pixels 3 and 6) is dark, but the text at pixel 3 (value 41) and the paper at pixel 8 (value 46) have almost the same value. The green line is the corrected output: paper is always 230 and text is always 51.

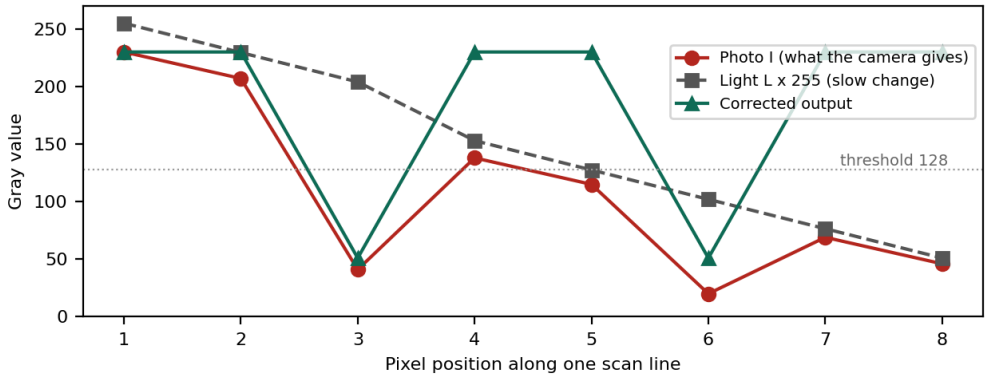


Figure 3: Red = photo, gray dashed = light, green = corrected output. The correction removes the slope and keeps the text.

5.4 The steps of homomorphic filtering

Step	What to do	Why / example
1	Take the gray photo I(x,y). Add 1 to every pixel.	Pixel = light x surface. Adding 1 avoids ln 0.
2	Take the natural log of every pixel: ln I.	Turns L x R into ln L + ln R.
3	Blur ln I with a large Gaussian filter to get ln L-hat (estimate of the light).	Light is slow (low frequency), so a big blur keeps only the light.
4	Subtract: ln R-hat = ln I - ln L-hat.	Slow light is removed, edges and text remain.
5	Take the exponential: R-hat = exp(ln R-hat).	Back to the normal brightness scale.
6	Stretch the result to 0 to 255 (normalise).	Output image is evenly lit and ready to use.

5.5 What the method does to our photo

Figure 4 shows a page with text. Panel 2 is the photo with a dark patch and light falling from left to right. After homomorphic filtering (panel 3) the dark patch and the left-to-right slope are almost gone, and the text is clear everywhere.

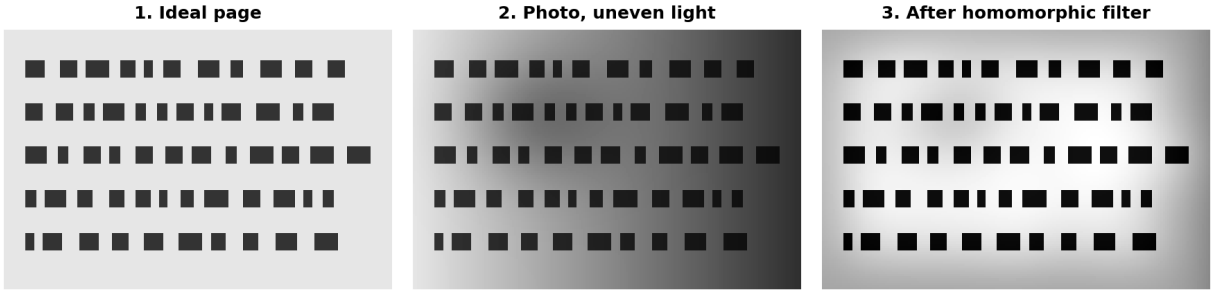


Figure 4: Ideal page, photo under uneven light, and result after homomorphic filtering (a slight fade at the extreme border is an edge effect).

5.6 Practical points

- **Blur size (sigma):** it must be much larger than the letters or details. If it is too small, the details are treated as light and removed (hollow, haloed letters). If it is too large, small patches of light are not removed.
- **Frequency-domain version:** take the Fourier transform of ln I, multiply by a high-frequency emphasis filter H(u,v) with gamma-L below 1 (reduces light) and gamma-H above 1 (boosts detail), then inverse transform and take exp. The steps above are the simple spatial version of the same idea.
- **Colour photos:** apply it only to the brightness channel (for example V in HSV or L in Lab) so the colours do not shift.
- **Border pixels:** the blur needs pixels outside the image. Replicate the nearest pixels (the usual choice) to avoid dark borders.

In OpenCV (Python) the whole method is a few lines:

```
import cv2, numpy as np
img   = cv2.imread('photo.png', 0)          # read as gray image
logi  = np.log(np.float32(img) + 1)         # step 1 and 2
light = cv2.GaussianBlur(logi, (0, 0), 50)  # step 3: estimate light
refl  = logi - light                        # step 4: remove light
out   = np.exp(refl)                        # step 5
out   = cv2.normalize(out, None, 0, 255, cv2.NORM_MINMAX).astype('uint8')
cv2.imwrite('even.png', out)                # step 6
```

6. Answer to (d): Why did you choose this method?

Answer: Uneven light is a slow change that is multiplied with the real picture. Homomorphic filtering is built for exactly this: it separates light from surface and removes only the light. Histogram equalisation and gamma correction use one mapping for the whole image, so they cannot fix bright and dark patches at the same time.

6.1 Same scan line, two methods

Use the 8-pixel scan line of Part 7. Both methods are judged by a simple test: a pixel above 128 is called paper (white), a pixel at or below 128 is called text (black). The real answer is: paper, paper, text, paper, paper, text, paper, paper.

Pixel	1	2	3	4	5	6	7	8
Real object	paper	paper	text	paper	paper	text	paper	paper
Photo value	230	207	41	138	115	20	69	46
Global threshold 128	white	white	black	white	**black (wrong)**	black	**black (wrong)**	**black (wrong)**
After homomorphic	230	230	51	230	230	51	230	230
Threshold 128 after correction	white	white	black	white	white	black	white	white

The global threshold (and any single global mapping) turns three paper pixels on the dark side black, because the dark side is dark everywhere. After homomorphic filtering the same threshold gives the correct answer for all 8 pixels.

6.2 Why not a global method?

A global method such as histogram equalisation replaces each gray value by a new value using **one table for the whole image**. In the example, pixel 3 (text, 41) and pixel 8 (paper, 46) are nearly equal, so any global table gives them nearly equal outputs. The text on the bright side stays as bright as the paper on the dark side. A method must look at the **local light** to fix this.

6.3 Comparison

Property	Global histogram equalisation / gamma	Homomorphic filtering
Idea	One mapping from old gray to new gray for all pixels	Separates light L from surface R, removes L
Uses position of the pixel	No	Yes (the light estimate changes from place to place)
Bright and dark patches together	Cannot fix both: brightens dark areas but washes out bright ones	Fixes both
Same surface, same output	No	Yes (paper becomes about the same value everywhere)
Edges and text	May be stretched with the noise	Kept, because they are high frequency
Type of filter	Point operation	Frequency / large-neighbourhood operation after log
Best used for	Image that is evenly lit but has low contrast	Image with slow, uneven lighting

Other options: CLAHE (adaptive histogram equalisation) and adaptive thresholding also work locally and are good alternatives, especially when the final goal is only black and white text. Homomorphic filtering is chosen here because it removes the cause (the light) and gives a corrected gray image that can be used for any later step.

7. Answer to (e): Step by step with a full example

Answer: Photo, add 1 and take log, blur to estimate light, subtract, take exp, normalise, evenly lit image.

7.1 Example input and output

We use one scan line of 8 pixels from a photo of a page. The paper has reflectance R = 0.9 and the text has R = 0.2. The text is at pixel 3 and pixel 6. The light L falls from 1.0 on the left to 0.2 on the right. Pixel value = 255 x L x R, rounded.

Pixel	1	2	3	4	5	6	7	8
Object	paper	paper	text	paper	paper	text	paper	paper
R (surface)	0.9	0.9	0.2	0.9	0.9	0.2	0.9	0.9
L (light)	1.0	0.9	0.8	0.6	0.5	0.4	0.3	0.2
Photo I = 255 x L x R	230	207	41	138	115	20	69	46

Notice that the paper changes from 230 to 46 and that the text at pixel 3 (41) is almost the same as the paper at pixel 8 (46). This is the problem to fix.

7.2 Detailed steps for one pixel (pixel 3, text, value 41)

Step 1: Take the photo value
I = 41 (a dark text pixel on the bright-left side). Add 1: 42.

Step 2: Take the log
ln I = ln 42 = 3.74. This equals ln L + ln R.

Step 3: Estimate the light (large blur)
The blur of the log values around this place gives ln L-hat = ln(0.8 x 255) = ln 204 = 5.32 (light is 0.8 here).

Step 4: Subtract the light
ln R-hat = 3.74 - 5.32 = -1.58. The slow light is removed; only the surface remains.

Step 5: Take the exponential
R-hat = exp(-1.58) = 0.21, which is close to the real reflectance 0.2 of the ink.

Step 6: Normalise to 0 to 255
Output = 0.21 x 255 = about 51. The text pixel is now 51, a dark value.

Figure 5: The steps for pixel 3. The pixel value 41 (which looked like paper on the dark side) becomes 51 (clearly text).

7.3 Repeat for all pixels

Do the same steps for every pixel with its own light estimate. For pixel 8 (paper, value 46, light 0.2): R-hat = 46 / (0.2 x 255) = 0.90, so the output is 0.90 x 255 = 230. The simple form of the steps is: **output = I / (255 x L-hat) x 255**. Figure 6 shows all 8 pixels.

Pixel	Old I	Estimated light L-hat	R-hat = I / (255 x L-hat)	New (R-hat x 255)
1	230	1.0	0.90	**230**
2	207	0.9	0.90	**230**
3	41	0.8	0.20	**51**
4	138	0.6	0.90	**230**
5	115	0.5	0.90	**230**
6	20	0.4	0.20	**51**
7	69	0.3	0.90	**230**
8	46	0.2	0.90	**230**

Figure 6: Calculation for all 8 pixels. Paper becomes 230 and text becomes 51 everywhere.

7.4 Check your result

- Before correction the paper changed from 230 to 46 (a drop of 184 gray levels). After correction every paper pixel is 230 and every text pixel is 51.
- Text and paper are now separated by a large gap (51 against 230), so one threshold of 128 works for the whole line.
- The corrected values match the real reflectance (0.9 and 0.2), so the picture looks as if it were evenly lit.

Note: in the example the light estimate is taken as known so that the numbers are easy to follow. In a real image it comes from the large Gaussian blur of the log image (Step 3), which gives a close, smooth estimate.

7.5 The complete process in one picture

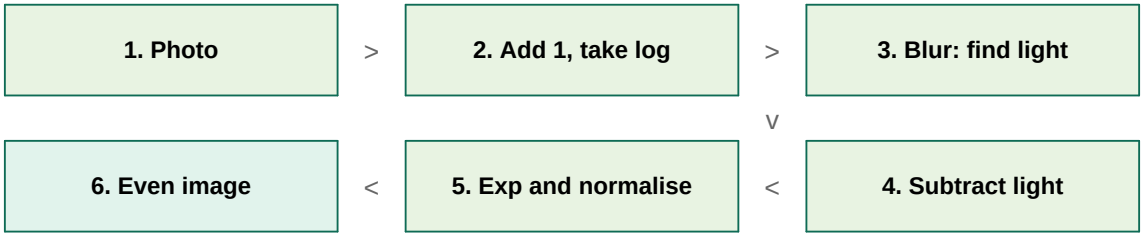


Figure 7: Flow of the homomorphic filtering process.

8. Short exam-ready answer

Part	Answer
(a)	Dark areas lose detail, bright areas are washed out, the same object looks different in different places, and thresholding, edge detection and OCR fail.
(b)	The light on the scene is not uniform. Causes: one-sided light source, shadows, mixed lights, lens vignetting, tilted or curved surface, limited camera exposure range.
(c)	Homomorphic filtering. Step 1: take the photo and add 1. Step 2: take the log, so I = L x R becomes ln L + ln R. Step 3: blur with a large Gaussian to estimate the light. Step 4: subtract the light. Step 5: take exp. Step 6: normalise to 0 to 255.
(d)	Light changes slowly and is multiplied with the real picture. Homomorphic filtering separates the two and removes only the light, so it fixes bright and dark patches together and keeps edges. Global histogram equalisation or gamma uses one mapping for all pixels and cannot do this.
(e)	Photo, add 1 and take log, blur to estimate light, subtract, exp, normalise, evenly lit image.
Example	Scan line: 230, 207, 41, 138, 115, 20, 69, 46 (text at pixels 3 and 6, light 1.0 down to 0.2). After correction: 230, 230, 51, 230, 230, 51, 230, 230. Threshold 128 now gives the correct text and paper everywhere.
Conclusion	Homomorphic filtering is the best choice for uneven illumination because it removes the slow light variation without blurring the edges and text.

9. Extra viva questions

Question	Short answer
Why do we take the log?	Because I = L x R is a product. The log turns it into a sum, ln L + ln R, so a filter can separate the two parts.
Is illumination low or high frequency?	Low frequency. It changes slowly. Edges and text (reflectance) are high frequency.
Why add 1 before the log?	ln 0 is not defined, and pixels can be 0. Adding 1 avoids this problem.
What happens if the blur is too small?	The details are treated as light and removed, so letters look hollow or have halos.
What happens if the blur is too large?	Small patches of light are not removed, so some unevenness remains.
Difference from histogram equalisation?	Histogram equalisation is one global mapping. Homomorphic filtering uses the light at each place, so it fixes local patches.
Which other methods can be used?	CLAHE (adaptive histogram equalisation), adaptive thresholding, and background subtraction (morphological top-hat or dividing by a blurred copy).
How is a colour photo corrected?	Apply the method only to the brightness channel (V in HSV or L in Lab) and keep the colour channels.
Which filter do we use in the frequency domain?	A high-frequency emphasis filter with gamma-L below 1 and gamma-H above 1.